

Section 10.1

In these exercises R is a ring with 1 and M is a left R -module.

Exercise (10.1.1). Prove that $0m = 0$ and $(-1)m = -m$ for all $m \in M$.

Proof: Suppose R is a ring with 1 and M is a left R -module. Since 0 is the identity under addition and since R modules are transitive we get $0m = (0+0)m = 0m + 0m$. Since we have elements of an abelian group on both sides we can use the cancellation law to get $0 = 0m$.

Note that since by definition we have that $1m = m$ we get that $m + (-1)m = 1m + (-1)m = (1 - 1)m = 0m = 0$ so in the abelian group M , m and $(-1)m$ are inverses, since inverses are unique we conclude that $(-1)m = -m$. $\square$

Exercise (10.1.8). An element m of the R -module M is called a torsion element if $rm = 0$ for some nonzero element $r \in R$. The set of torsion elements is denoted,

$$\text{Tor}(M) = \{m \in M \mid rm = 0 \text{ for some nonzero } r \in R\}$$

Proof: (a) Prove that if R is an integral domain then $\text{Tor}(M)$ is a submodule of M (called the torsion submodule of M).

Proof: Suppose R is an integral domain. Clearly $0 \in \text{Tor}(M)$ since $r0 = 0$ for all $r \in R$. Let $x, y \in \text{Tor}(M)$ and $r \in R$. By definition there exists some nonzero $r_x, r_y \in R$ such that $r_x x = 0$ and $r_y y = 0$. Consider $x + ry$, and note that since R is an integral domain it is commutative so multiplying on the left by $r_x r_y$ gives the following,

$$r_x r_y (x + ry) = (r_x r_y)x + (r_x r_y)ry = r_y(r_x x) + r_x r(r_y y) = r_y(0) + r_x r(0) = 0.$$

Since $r_x r_y \in R$ we know that $x + ry \in \text{Tor}(M)$ and therefore by The Submodule Criterion $\text{Tor}(M)$ is a submodule of M . $\square$

- (b) Give an example of a ring R and an R -module M such that $Tor(M)$ is not a submodule.

Proof:

□

- (c) If R has zero divisors show that every nonzero R -module has nonzero torsion elements.

Proof:

□

□

Exercise (10.1.9). If N is a submodule of M , the annihilator of N in R is defined to be $\{r \in R \mid rn = 0 \text{ for all } n \in N\}$. Prove that the annihilator of N in R is a 2-sided ideal of R .

Proof: Suppose N is a submodule of M let $ann(N)$ be the annihilator of N in R . Let $a, b \in ann(N)$ and $n \in N$, now note that $(a - b)n = an - bn = 0 - 0 = 0$. Therefore $ann(N)$ is a subring of R . Let $a \in ann(N)$ and $r \in R$, consider $n \in N$, therefore $n(ar) = (na)r = 0r = 0$ so $ar \in ann(N)$. Now note that $(ra)n = r(an) = r0 = 0$ so $ra \in ann(N)$. Therefore $ann(N)$ is a 2-sided ideal of R .

□

Exercise (10.1.10). If I is a right ideal of R , the annihilator of I in M is defined to be $\{m \in M \mid am = 0 \text{ for all } a \in I\}$. Prove that the annihilator of I in M is a submodule of M .

Proof: Suppose I is a right ideal of R . Clearly $0 \in ann(I)$ since for all $a \in I$, $a(0) = 0$ so $ann(I) \neq \emptyset$. Let $x, y \in ann(I)$ and note that for all $a \in I$ and $r \in R$ we get $a(x + ry) = ax + ary$ since I is a right ideal $ar \in I$ and therefore we get $a(x + ry) = ax + ary = 0 + 0 = 0$. So $x + ry \in ann(I)$ and thus by the submodule criterion, $ann(I)$ is a submodule of M .

□

Exercise (10.1.11). Let M be the abelian group $\mathbb{Z}/24\mathbb{Z} \times \mathbb{Z}/15\mathbb{Z} \times \mathbb{Z}/50\mathbb{Z}$.

- (a) Find the annihilator of M in $\mathbb{Z}$ (i.e, a generator for this principal ideal).

Proof: Consider the ideal generated by the $l = \text{lcm}(24, 15, 50)$. We will proceed by showing that $(l) = \text{ann}_{\mathbb{Z}}(M)$. Suppose $a \in (l)$ so $a = lj$ some j multiple of l in $\mathbb{Z}$ and let $m \in M$ such that $m = (a, b, c)$ where $a \in \mathbb{Z}/24\mathbb{Z}$, $b \in \mathbb{Z}/15\mathbb{Z}$ and $c \in \mathbb{Z}/50\mathbb{Z}$. Then note that since $l = \text{lcm}(24, 15, 50)$ we get $am = (lja, ljb, ljc) = (0, 0, 0)$. So $a \in \text{ann}_{\mathbb{Z}}(M)$ and $(l) \subseteq \text{ann}_{\mathbb{Z}}(M)$.

Let $j \in \text{ann}_{\mathbb{Z}}(M)$. Then by definition for all $m = (a, b, c) \in M$ we get that $jm = (ja, jb, jc) = 0$. So it must follow that that $j(1, 1, 1) = (j, j, j) = (0, 0, 0)$ so it must be the case that $j \in (\text{lcm}(24, 15, 50)) = (l)$. Therefore $\text{ann}_{\mathbb{Z}}(M) \subseteq (l)$, thus finally $(l) = \text{ann}_{\mathbb{Z}}(M)$ □

- (b) Let $I = 2\mathbb{Z}$. Describe the annihilator of I in M as a direct product of cyclic groups.

Proof:

□

Section 10.2

In these exercises R is a ring with 1 and M is a left R -module.

Exercise (10.2.1). Use the submodule criterion to show that kernels and images of R -module homomorphisms are submodules.

Proof: Suppose that R is a ring with 1 and M and N be R -modules with the map $\varphi : M \rightarrow N$ being an R -module homomorphism. We want to show that $\ker(\varphi)$ is a submodule of M and $\varphi(M)$ is a submodule of N . Note that $\ker(\varphi) \neq \emptyset$ since $1 \in \ker(\varphi)$. Let $x, y \in \ker(\varphi)$ and $r \in R$. Note that since φ is an R -module homomorphism we know

that $\varphi(x + ry) = \varphi(x) + r\varphi(y) = 0 + r0 = 0$. So $x + ry \in \ker(\varphi)$ and therefore by the submodule criterion $\ker(\varphi)$ is a submodule.

Now let $x, y \in \varphi(M)$ and $r \in R$. Note that if $x, y \in \varphi(M)$ then there exists an $a, b \in M$ such that $\varphi(a) = x$ and $\varphi(b) = y$. Since φ is an R -module homomorphism we know that $\varphi(a + rb) = \varphi(a) + r\varphi(b) = x + ry$ and so $x + ry \in \varphi(M)$. By the submodule criterion we know that $\varphi(M)$ is a submodule. $\square$

Exercise (10.2.2). Show that the relation 'is R -module isomorphic to' is an equivalence relation on any set of R -modules.

Proof:

$\square$